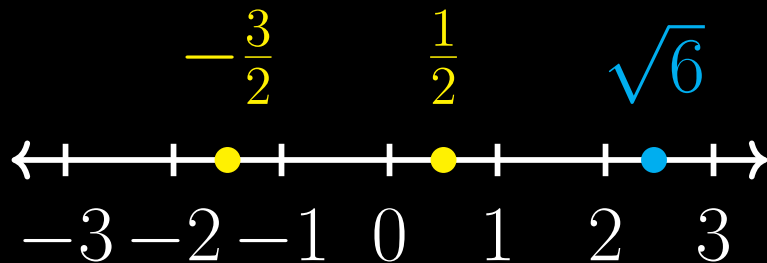


Subsets of Real Numbers

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The set of **Real Numbers** ($\mathbb{R}$) consists of all numbers that can be located on the number line.

Real Numbers on the Number Line



Every point on the number line is a real number.

Natural Numbers

The set of **Natural Numbers** ($\mathbb{N}$) is

$$\mathbb{N} = \{1, 2, 3, \dots\}$$

These are the numbers we use for counting.

Whole Numbers

The set of **Whole Numbers** ($\mathbb{W}$) is

$$\mathbb{W} = \{0, 1, 2, 3, \dots\}$$

Whole numbers are the natural numbers together with 0.

Integers

The set of **Integers** ($\mathbb{Z}$) includes negative whole numbers, zero, and positive whole numbers.

$$\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$$

The set of **Rational Numbers** ($\mathbb{Q}$) consists of numbers that can be written as $\frac{a}{b}$ where a and b are integers and $b \neq 0$.

Rational Number Examples

$$8 = \frac{8}{1} \quad 0.5 = \frac{1}{2} \quad -\frac{3}{5}$$

Any integer n can be written as $\frac{n}{1}$, so every integer is rational.

Decimals of Rational Numbers

The decimal form of a rational number either **terminates** or **repeats**.

$$0.25 = \frac{1}{4} \qquad 0.666 \dots = \frac{2}{3}$$

The set of **Irrational Numbers** consists of numbers that cannot be written as a ratio of two integers. Their decimal forms are non-terminating and non-repeating.

Irrational Number Examples

$$\sqrt{6} = 2.449489 \dots$$

$$\sqrt{8} = 2.828427 \dots$$

The digits go on forever without a repeating pattern.

Square Roots

The square root of a perfect square is rational.

$$\sqrt{25} = 5$$

The square root of a whole number that is not a perfect square is irrational.

$$\sqrt{6} \text{ is irrational}$$

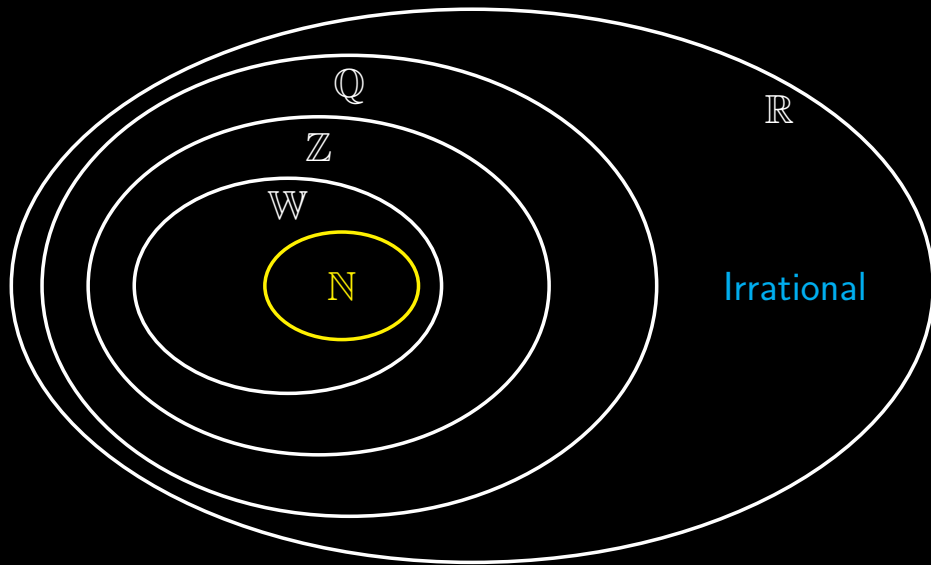
Every natural number is also a whole number, every whole number is an integer, every integer is a rational number, and every rational number is a real number.

Chain of Subsets

The subsets of real numbers can be illustrated using a nested diagram:

$$\mathbb{N} \subset \mathbb{W} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$$

Nested Diagram



Where Numbers Belong

	N	W	Z	Q	Irr.	R
8	✓	✓	✓	✓		✓
0		✓	✓	✓		✓
-5			✓	✓		✓
$\frac{3}{4}$				✓		✓
$\sqrt{6}$					✓	✓

Memory Tip

**No Wonder Zebras Quietly
Run**

N atural	W hole	Z (Integers)	Q (Ratio-
nal)	R real		

Examples

Example 1

Classify each number as **Natural**, **Whole**, **Integer**, **Rational**, or **Irrational**.

(a) 25 **Natural**

(b) -12 **Integer**

(c) $\frac{3}{7}$ **Rational**

(d) $\sqrt{6}$ **Irrational**

Example 2

Write all the subsets of real numbers to which each number belongs.

(a) 15: $\mathbb{N}$, $\mathbb{W}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$

(b) -3 : $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$

(c) $\frac{2}{3}$: $\mathbb{Q}$, $\mathbb{R}$

(d) $\sqrt{8}$: Irrational, $\mathbb{R}$

Example 3

Identify whether each statement is **True** or **False**.

(a) Every integer is a whole number. **False**,
since -3 is an integer but not a whole number.

(b) Every natural number is a real number.
True

(c) No irrational number is rational. **True**

Example 4

Classify each decimal number.

(a) 0.5 Rational (terminates)

(b) $0.272727 \dots$ Rational (repeats)

(c) $2.828427 \dots$ Irrational (non-terminating
and non-repeating)

Example 5

Which set best describes each number?

(a) $\sqrt{16} = 4$ Natural

(b) $-\sqrt{25} = -5$ Integer

(c) $\frac{-12}{4} = -3$ Integer

(d) $\sqrt{12}$ Irrational

(e) $\frac{5}{9}$ Rational

Example 6

Which numbers belong to the set of integers?

$$-14, \frac{3}{8}, 6, \sqrt{6}, 0$$

The integers are -14 , 6 , and 0 .

Example 7

Which numbers are irrational?

$$\sqrt{16}, \frac{3}{5}, \sqrt{6}, 7, \sqrt{18}$$

The irrational numbers are $\sqrt{6}$ and $\sqrt{18}$.

Practice Exercises

Practice 1

Classify each number as **Natural**, **Whole**, **Integer**, **Rational**, or **Irrational**.

(a) 9

(b) 0

(c) -6

(d) $\frac{5}{8}$

(e) $\sqrt{2}$

Practice 2

Write all the subsets of real numbers to which each number belongs.

(a) 12 (b) 0 (c) -9

(d) $\frac{7}{4}$ (e) π

Practice 3

Identify whether each statement is **True** or **False**.

- (a) Every whole number is an integer.
- (b) Every rational number is an integer.
- (c) Every irrational number is a real number.
- (d) Zero is a natural number.
- (e) Every integer is rational.

Practice 4

Classify each decimal number.

(a) 0.75

(b) -3.2

(c) $0.121212\dots$

(d) $1.414213\dots$

(e) 8.0

Practice 5

Which set best describes each number?

(a) $\sqrt{36}$ (b) $-\sqrt{49}$ (c) $\frac{-10}{5}$

(d) $\sqrt{5}$ (e) $\frac{11}{2}$

Practice 6

Which numbers belong to the set of integers?

$$-8, 0, \frac{2}{3}, 15, \sqrt{7}, -21$$

Practice 7

Which numbers are irrational?

$$\sqrt{11}, \frac{9}{10}, \pi, 4, \sqrt{49}, \sqrt{13}$$

Activities

Activity 1

Classify each number as **Natural**, **Whole**, **Integer**, **Rational**, or **Irrational**.

(a) 14

(b) 0

(c) -15

(d) $\frac{9}{11}$

(e) $\sqrt{3}$

Activity 2

Write all the subsets of real numbers to which each number belongs.

(a) 20 (b) 0 (c) -18

(d) $\frac{13}{5}$ (e) $\sqrt{17}$

Activity 3

Identify whether each statement is **True** or **False**.

- (a) Every natural number is a whole number.
- (b) Every irrational number is rational.
- (c) Every integer is a real number.
- (d) Zero is a whole number.
- (e) Every rational number is real.

Activity 4

Classify each decimal number.

(a) 0.625

(b) -7.4

(c) $0.454545\dots$

(d) $3.141592\dots$

(e) 11.0

Activity 5

Which set best describes each number?

(a) $\sqrt{81}$ (b) $-\sqrt{64}$ (c) $\frac{-24}{6}$

(d) $\sqrt{19}$ (e) $\frac{17}{3}$

Activity 6

Which numbers belong to the set of integers?

$$-30, 8, \frac{5}{6}, 0, \sqrt{15}, 27$$

Activity 7

Which numbers are irrational?

$$\sqrt{10}, \frac{4}{7}, \pi, \sqrt{64}, \sqrt{21}, 9$$

Thank you!